What a causal mask is worth: a rank–logit law for prefix averaging

Abstract

A causal mask realizes prefix averaging with zero learned scores. We measure the matrix-level resources used by a position-dependent unmasked softmax head at one fixed sequence length. For row-centered score rank r, the minimum centered logit radius at worst-row output error n^{-p}, with fixed $p > 1$ and fixed r, satisfies

$$\log B = \frac{n-1}{r}\big((p-1)\log n + \log\log n + O_{p,r}(1)\big).$$

The formula resolves both diverging terms in the fixed-rank exponent at every fixed polynomial target. The proof expands a determinant for the lower bound and builds a cost-balanced rank-r matrix for the upper bound. A companion estimate uniform in rank forces $\Omega(n)$ rank under a polynomial radius budget when $p > 3/2$. A factor-norm inequality transfers the radius requirement to dot-product factors. In a range-only exact-real model, the finite float32 factor range excludes the $p = 2$ target for position-indexed pure dot-product scores at head dimension 128 once $n \geq 4166$. The same exact-real, float32-range assumptions require ambient head dimension above 252 at $n = 8192$. A finite-size audit places the construction between a row-weighted determinant bound and its closed-form budget. A content-adaptive extension holds on Boolean payloads: when the scores may read the averaged bits through a linear channel, the all-zero realization is forced entrywise close to causal prefix attention, and the fixed-rank law is unchanged. Every other statement concerns one fixed pure dot-product attention matrix and holds uniformly over bounded value streams while that matrix stays fixed.

1 Introduction

For prefix averaging, a causal mask supplies a moving support boundary while keeping every visible score flat. Let

$$T(i,j) = \frac{1}{i}\mathbf{1}[j \leq i]$$

be the prefix-average operator. A masked head with every visible score equal to zero realizes T exactly. An unmasked softmax assigns positive mass to every key. Accurate realization requires nearly equal scores on each prefix together with strong suppression of every future key. Low score rank makes that joint cost accumulate across positions.

For an unmasked score matrix S, let $P_0 = I_n - \mathbf{1}\mathbf{1}^\top/n$. The centered score rank is $\rho(S) = \operatorname{rank}(SP_0)$ and centered logit radius $B(S) = \max_{i,j} |(SP_0)(i,j)|$. For a dot-product head of dimension d_h, $\rho(S) \leq d_h$. Any matrix with $\rho(S) \leq r$ admits an r-dimensional factorization because $\operatorname{softmax}(S) = \operatorname{softmax}(SP_0)$. Equal finite row-softmax matrices have logits that differ by rowwise constants, since their pairwise log-probability ratios agree. Thus ρ is the minimum factor dimension after quotienting row shifts. At every fixed $p > 1$ and fixed r, the main theorem gives

$$r \log B(S) = (n-1)\big((p-1)\log n + \log\log n + O_{p,r}(1)\big)$$

at the optimum for error n^{-p}. For a position-dependent dot-product head with fixed dimension d_h, the required centered score magnitude grows as $\exp(\Theta(n \log n/d_h))$. A fixed polynomial score budget requires dimension growing linearly with sequence length when $p > 3/2$. The excluded lengths in Section 5 sit one to two orders of magnitude below deployed decoder contexts.

The result concerns one fixed, length-specific pure dot-product matrix evaluated uniformly over bounded value streams. Section 6 extends the fixed-rank law to scores that adapt to a Boolean payload. Section 8 states the remaining model boundaries and collects the open questions.

The table compares the realizations at a common accuracy ε. Its third row uses $\varepsilon = n^{-p}$ for fixed $p > 1$, where the two-term law applies. The masked row uses zero learned scores. Its architectural support constraint supplies the forbidden entries and falls outside the score charge.

realization	accuracy	centered score rank	realized centered-logit radius
causally masked	any ε	0	0
unmasked, full centered rank	any ε	$n-1$	$O(\log(n/\varepsilon))$
unmasked, fixed rank r	$\varepsilon = n^{-p}$	r	$\exp(\Theta(n\log n/r))$

The lower-bound proof takes first differences across all causal boundaries, producing a square matrix whose strict lower triangle is small, whose diagonal is large, and whose rank is at most r. A columnwise determinant expansion removes every term that uses more than r columns from this matrix, giving the factor $(n-1)/r$ that fixed $(r+1)$-minors miss. Because the argument works on principal submatrices, it can omit inaccurate rows.

The upper bound assigns the nonterminal rows to r consecutive blocks. Score drops grow geometrically inside each block, with block boundaries chosen to balance their logarithmic cost. The lower and upper bounds agree through $(p-1)\log n$ and $\log\log n$ in the exponent.

2 Operational model

Let $S(i,j)=\langle q_i,k_j\rangle$ and $A=\mathrm{softmax}_{\mathrm{row}}(S)$. We take S to be the entire pre-softmax score matrix. Any additive positional-bias matrix belongs in S and in its rank and radius charges. The conventional $1/\sqrt{d_h}$ scale can be absorbed into the factors. Softmax is invariant under row shifts, so SP_0 is the identifiable score matrix. Its radius controls every within-row score difference:

$$|S(i,j)-S(i,t)|\le 2B(S).$$

For a bounded value stream v, the output is Av. The operational error is

$$\sup_{\|v\|_\infty\le 1}\|(A-T)v\|_\infty=\max_i\|A(i,:)-T(i,:)\|_1. \tag{1}$$

Hölder's inequality gives the upper bound, and the sign vector of the largest-error row gives equality. The metric asks one realized attention matrix to average every bounded value stream.

Uniformity ranges over separate value streams while A stays fixed. A single fixed value matrix tests the column span that it presents to the head.

The radius $B(S)$ also admits an architectural bound.

Proposition 1 (Factor-norm bound). *Let $S=QK^\top$ with rows $q_i,k_j\in\mathbb{R}^{d_h}$, and let $\bar k=\frac{1}{n}\sum_j k_j$. Then*

$$B(S)\le\max_i\|q_i\|_2\cdot\max_j\|k_j-\bar k\|_2. \tag{2}$$

If in addition every entry of Q and K has magnitude at most F, then $\log B(S)\le\log 2+\log d_h+2\log F$.

Proof. Row i of SP_0 has entries $\langle q_i,k_j\rangle-\frac{1}{n}\sum_t\langle q_i,k_t\rangle=\langle q_i,k_j-\bar k\rangle$, and Cauchy–Schwarz gives (2). Under the entrywise bound, $\|q_i\|_2\le\sqrt{d_h}F$ and $\|k_j-\bar k\|_2\le 2\sqrt{d_h}F$. ☐

The lower bounds force a product of query and centered-key norm envelopes. A finite exponent range places an additive cap on $\log B(S)$. Under the conventional $1/\sqrt{d_h}$ scaling the right-hand side of (2) is divided by $\sqrt{d_h}$.

Write $[m]=\{1,\dots,m\}$. For a finite set, osc denotes the maximum minus the minimum, and $\left\{{m \atop k}\right\}$ denotes a Stirling number of the second kind. Let $I\subseteq[n-1]$ and put

$$I^+=I\cup\{j+1:j\in I\}.$$

Row $i\in I$ has a causal (δ_i,γ_i) profile *on I* when

$$\mathrm{osc}_{j\in I^+,\ j\le i}\,S(i,j)\le\delta_i,\qquad S(i,i+1)\le S(i,i)-\gamma_i. \tag{3}$$

Accuracy sets explicit row-dependent parameters.

Lemma 2 (Output error gives score geometry). *If $\max_j|A(i,j)-T(i,j)|\le\varepsilon$ and $i\varepsilon<1/2$, then row i satisfies (3) on every I, with*

$$\delta_i=\log\frac{1+i\varepsilon}{1-i\varepsilon},\qquad \gamma_i=\log\frac{1-i\varepsilon}{i\varepsilon}. \tag{4}$$

The hypothesis uses the entrywise matrix error $\max_{i,j}|A(i,j)-T(i,j)|\le\varepsilon$. It follows from the operational row-ℓ_1 metric $\|A(i,:)-T(i,:)\|_1\le\varepsilon$ in (1). Section 4 constructs a matrix that attains the row-ℓ_1 target. The norm sandwich $\|x\|_\infty\le\|x\|_q\le\|x\|_1$ establishes the same two-term law for maximum row-ℓ_q error at every fixed $1\le q\le\infty$. In probe language, entrywise matrix error corresponds to value streams with $\|v\|_1\le 1$.

The scale $1/n$ has a direct interpretation. For a binary value stream, $i(Tv)_i$ is an integer prefix sum. Rounding $i(Av)_i$ recovers that sum whenever the scalar output error is below $1/(2i)$. The eligibility condition $i\varepsilon<1/2$ marks this recovery scale. Fixed targets n^{-p}, $p>1$, give every nonterminal boundary a polynomial margin.

3 The rank–logit law

The finite inequality below supplies the matrix-theoretic core.

Theorem 3 (Row-subset determinant bound)**.** *Let $\varnothing \neq I \subseteq [n-1]$, $m = |I|$, and suppose the rows in I satisfy (3) on I, with positive δ_i, γ_i. If $1 \le r$, $\rho(S) \le r$ and $B(S) \le B$, then*

$$\prod_{i\in I} \gamma_i \le \sum_{k=1}^{\min\{r,m\}} k! \left\{ {m \atop k} \right\} (2B)^k \max_{\substack{R\subseteq I \\ |R|=k}} \prod_{i\in I\setminus R} \delta_i. \tag{5}$$

If additionally $\delta_i \le \delta$ and $\gamma_i \ge \gamma$ on I, then

$$\gamma^m \le \sum_{k=1}^{\min\{r,m\}} k! \left\{ {m \atop k} \right\} \delta^{m-k} (2B)^k. \tag{6}$$

The theorem uses the eligible rows in I, meaning rows with $i\varepsilon < 1/2$. Accuracy on αn such rows replaces the determinant size $n-1$ by αn, up to rounding; a corresponding uniform asymptotic law also requires their δ_i, γ_i to have uniform margins. When $\varepsilon = o(1/n)$, every nonterminal row is eligible, and average row error at most $\varepsilon/2$ supplies at least $n/2 - 1$ accurate nonterminal rows by Markov's inequality, up to rounding. The determinant argument uses only this rowwise consequence.

For $0 < \varepsilon < 1$ and $1 \le r \le n-1$, define

$$B_r^{\text{out}}(n,\varepsilon) = \inf \left\{ B(S) : \rho(S) \le r, \sup_{\|v\|_\infty \le 1} \|(A-T)v\|_\infty \le \varepsilon \right\}, \qquad B_{r,p}^{\text{out}}(n) = B_r^{\text{out}}(n, n^{-p}).$$

Write $N = n-1$ through the rest of this section.

Theorem 4 (Cost-balanced rank-r construction)**.** *Let $n \ge 2$, $0 < \varepsilon \le 1/n$, $N = n-1$, and $1 \le r \le N$. There is $S \in \mathbb{R}^{n\times n}$ with $\operatorname{rank}(S) = \rho(S) = r$, output error at most ε, and*

$$\log B(S) \le \frac{N}{r} \log \frac{64e \log(8n/\varepsilon)}{n\varepsilon} + \log\log \frac{8n}{\varepsilon}. \tag{7}$$

Theorem 5 (Two-term fixed-rank output law)**.** *For every fixed $p > 1$ and fixed positive integer r,*

$$\log B_{r,p}^{\text{out}}(n) = \frac{n-1}{r} \big((p-1)\log n + \log\log n + O_{p,r}(1) \big). \tag{8}$$

Proof. Set $\varepsilon = n^{-p}$. For all sufficiently large n, every nonterminal row is eligible in Lemma 2, and

$$\delta_i \le \delta = 2n^{1-p}(1+o(1)), \qquad \gamma_i \ge \gamma = (p-1)\log n + o(1).$$

Two entries in a profiled row differ by at least γ, so $2B \ge \gamma$. Eventually $\gamma \ge \delta$ and $N \ge r$. Apply Theorem 3 with $I = [N]$. The estimates $k!\left\{ {N \atop k} \right\} \le k^N \le r^N$ give

$$\gamma^N \le r^{N+1} \delta^{N-r} (2B)^r, \qquad \log B \ge \frac{N}{r} \log \frac{\gamma}{r\delta} + \log \frac{\delta}{2} - \frac{1}{r} \log r.$$

Here

$$\log \frac{\gamma}{r\delta} = (p-1)\log n + \log\log n + O_{p,r}(1).$$

The remaining term $\log(\delta/2) = O_p(\log n)$ fits inside $(N/r)O_{p,r}(1)$, which proves the required lower estimate.

Theorem 4 supplies a score matrix whose output error is at most ε and whose radius obeys

$$\log B \le \frac{N}{r} \log \frac{64e\log(8n/\varepsilon)}{n\varepsilon} + \log\log \frac{8n}{\varepsilon}.$$

Substitution of $\varepsilon = n^{-p}$ turns the logarithm multiplied by N/r into $(p-1)\log n + \log\log n + O_p(1)$. Its final additive term is $O_p(\log\log n)$, which also fits inside $(N/r)O_{p,r}(1)$. ☐

The $O_{p,r}(1)$ term permits a multiplicative $\exp(O(n/r))$ uncertainty in B. [1]

[1] For growing r, the permutation count in (6) contributes $-((n-1)/r)\log r$. Appendix C gives a complementary Hadamard bound that removes $\log r$ at a cost of $\frac{1}{2}\log n$, together with the all-rank floor $B \ge \max_i \gamma_i/2$.

Corollary 6 (Head-dimension tradeoff)**.** *Fix $p > 1$ and d_h. A position-dependent unmasked pure dot-product head of dimension d_h that reaches error n^{-p} obeys, for some constant C_{p,d_h} and all sufficiently large n,*

$$d_h \log B(S) \geq (n-1)\big((p-1)\log n + \log\log n - C_{p,d_h}\big).$$

The fixed-rank remainder depends on r. After normalization by N/r, the budget in (7) equals $(p-1)\log n + \log\log n + O_p(1)$ uniformly for $r = o(N/\log\log n)$. The determinant count introduces $-\log r$ into the lower estimate. Both diverging terms continue to match when $\log r = o(\log\log n)$; equation (8) states the asymptotic with r fixed. At full centered rank, Proposition 8 attains $\log B = \log\log(n/\varepsilon) + O(1)$. At $r = N$, the closed-form budget (7) gives $(p-1)\log n + 2\log\log n + O_p(1)$, and the singleton-block realization itself has $\log B = \log\log n + O_p(1)$. Appendix C develops the growing-rank estimate.

Corollary 7 (Polynomial radius budget)**.** *Fix $p > 3/2$ and $c > 0$. A score matrix with $B(S) \leq n^c$ and output error at most n^{-p} has*

$$\rho(S) \geq \left(\frac{p-3/2}{c+p-1} - o(1)\right) n.$$

The present uniform-in-rank Hadamard estimate yields a positive coefficient for $p > 3/2$. Behavior at the threshold remains open.

The finite inequality separates the accuracy regimes more precisely than a single asymptotic formula.

accuracy ε	eligible rows	status for fixed r
n^{-p}, fixed $p > 1$	all nonterminal	two-term law, Theorem 5
general $o(1/n)$	all nonterminal	two bounds with unmatched leading terms
c/n, fixed $c > 0$	linear subset	$\Omega_{c,r}(n/r)$ to $O_{c,r}(n\log\log n/r)$
$1/n \ll \varepsilon \ll 1$	$\Theta(1/\varepsilon)$	lower bound; construction open

At $\varepsilon = 1/(n\log n)$, the determinant bound gives $(N/r)(\log\log n + \log\log\log n + O(1))$. The budget in (7) gives $(N/r)(2\log\log n + O(1))$. Their leading coefficients differ, which restricts (8) to polynomial targets. For $\varepsilon = c/n$ with fixed $c > 0$, a doubly logarithmic gap remains. At larger ε, Lemma 2 certifies $\Theta(1/\varepsilon)$ rows. That count records the reach of the current proof. The behavior of B_r^{out} across the $1/n$ scale remains an open question.

Full centered rank changes the scaling.

Proposition 8 (Full-rank two-level scores)**.** *For $\Gamma > 0$, set*

$$S_\Gamma(i,j) = -\Gamma\,\mathbf{1}[j > i].$$

Then $\operatorname{rank}(S_\Gamma P_0) = n-1$, $B(S_\Gamma) \leq \Gamma$, and every nonterminal row has a causal $(0,\Gamma)$ profile on every I. Taking $\Gamma = \log(2(n-1)/\varepsilon)$, for $0 < \varepsilon \leq 1$, gives output error at most ε.

Proof. The submatrix on rows $1,\ldots,n-1$ and columns $2,\ldots,n$ is upper triangular with diagonal $-\Gamma$, so $\operatorname{rank}(S_\Gamma) \geq n-1$. Column 1 of S_Γ is zero, since $1 > i$ fails for every $i \geq 1$, so $e_1 \in \ker S_\Gamma$, the rank is exactly $n-1$, and $\ker S_\Gamma = \operatorname{span}\{e_1\}$. If $S_\Gamma P_0 x = 0$, then $P_0 x$ lies in this kernel and has coordinate sum zero, which gives $P_0 x = 0$ and $\ker(S_\Gamma P_0) = \operatorname{span}\{\mathbf{1}\}$. The centered rank is $n-1$. Row centering keeps every entry within the raw row range Γ. The future leakage in every row is at most $(n-1)e^{-\Gamma} \leq \varepsilon/2$; the prefix conditional distribution is uniform, so the row-ℓ_1 error is at most ε. □

The matrix S_Γ gives a finite-Γ approximation to the causal mask. It uses full centered rank and a growing realized radius to approximate the architectural support boundary.

4 Determinant mechanism and matching construction

Let $N = n-1$, let $\Delta \in \mathbb{R}^{n\times N}$ be the first-difference operator, and take the first N rows of $S\Delta$:

$$D(i,j) = S(i,j+1) - S(i,j).$$

Since $P_0\Delta = \Delta$, $\operatorname{rank}(D) \leq \rho(S)$. Row i's profile gives a small strict lower triangle and a large diagonal:

$$|D(i,j)| \leq \delta_i \quad (j < i), \qquad D(i,i) \leq -\gamma_i.$$

Let U be the upper-triangular part of D and $E = D - U$. Then $|\det U| \geq \prod_i \gamma_i$. Expand $\det(D-E)$ by columns. A term using more than r columns of D vanishes by rank. Summed over all k-subsets K of columns drawn from D, the number of surviving pairs (K, π) is $k!\left\{ {N \atop k} \right\}$; the count for an individual K is at most this and depends on K. Each term uses k entries bounded by $2B$ and $N-k$ lower-triangular entries. Keeping their row indices yields (5). Taking a principal submatrix $D[I, I]$ proves the row-subset form. Appendix B supplies the complete counting argument.

The construction balances row budgets over r consecutive blocks. One outer product creates the score drops inside each block. Appendix D gives the factorization and verifies its centered rank, row error, and radius bound (7).

5 Numerical checks and factor range

The audit evaluates the explicit construction and the finite determinant inequality at $p = 2$. High-precision decimal arithmetic computes the row-ℓ_1 error and centered radius. The lower curve solves (5) with the row profile from Lemma 2 and the all-rank floor.

Every quantity in the finite inequality is measurable on a realized score matrix: per-row oscillations and drops, the centered radius, and the factor dimension can be read off logits directly, so (5) evaluates at any (n, r, ε) with no asymptotic step. The test is one-directional: violation excludes accuracy ε for the measured resources; satisfaction certifies nothing.

For a radius value B, define its normalized remainder at $p = 2$, the value used throughout this section, by

$$R_{n,r}(B) = \frac{r}{n-1} \log B - \big(\log n + \log\log n\big),$$

the general leading term being $(p-1)\log n + \log\log n$. Theorem 5 predicts $R_{n,r} = O_r(1)$ for fixed r. Figure 1 places the construction between the finite lower bound and its closed-form budget in all 36 records. Every measured row error is below $0.251\, n^{-2}$. The displayed interval bounds the unknown optimum.

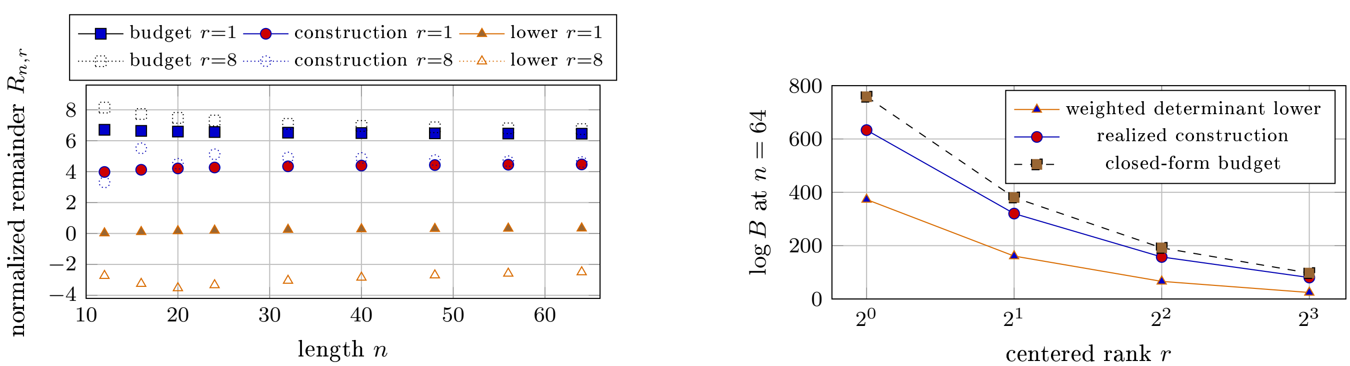

Figure 1: Finite-size construction and bracket at $p = 2$. Left: the three normalized remainders as n varies, with $r = 1$ solid and $r = 8$ dotted. At $n = 64$, the full bracket spans 6.12 nats for $r = 1$ and 9.25 for $r = 8$. Right: the row-weighted determinant lower bound, realized construction, and closed-form budget at $n = 64$.

5.1 A representability corollary

Proposition 1 converts the finite lower bound into a statement about every exact-real head with bounded factor entries.

Corollary 9 (Representable factors)**.** *Fix $p > 1$ and a floating-point format whose largest finite magnitude is F. Let $Q, K \in \mathbb{R}^{n \times d_h}$ have entries bounded in magnitude by F, and evaluate $QK^\top$ and softmax over the exact reals. Worst-row error at most n^{-p} requires the finite inequality (5), with $r = d_h$, to admit $\log B \le \log 2 + \log d_h + 2\log F$. Failure of this condition excludes such a head.*

The floating-point format supplies a dynamic-range bound on exact-real factors. Evaluation of (5) at $p = 2$ gives the table. Each entry is the largest n admitted by the finite bound. The bfloat16 and float32 thresholds are computed from distinct exact maxima ($\log F = 88.7189$ and 88.7228); both round to the displayed value, and the admitted lengths coincide.

format	$\log F$	$d_h = 8$	32	64	128
float16	11.09	65	262	531	1075
bfloat16	88.72	268	1053	2093	4165
float32	88.72	268	1053	2093	4165

The first failing length remains excluded as n grows because each δ_i decreases and each γ_i increases on the same eligible row subset. At $n = 8192$, the finite inequality gives $\log B \geq 418.72$ for $r = 128$. Proposition 1 then forces

$$\max_i \|q_i\|_2 \max_j \|k_j - \bar{k}\|_2 \geq e^{418.72}.$$

The same calculation excludes ambient dimensions through 252 for float32 or bfloat16 and through 934 for float16. Every claim in this paragraph uses exact-real evaluation with the stated factor envelope.

For scale, deployed decoder contexts run from 1.3×10^5 to beyond 10^6 positions [7, 11]; every admitted length in the table lies below 4.2×10^3, and the anchor length $n = 8192$ is a sixteenth of the shorter figure.

6 Content-adaptive scores

The bounds so far hold one score matrix fixed while value streams vary. This section removes that restriction for a Boolean payload class: the scores may read the payload they average, and the fixed-rank law is unchanged.

Position j carries a payload bit $v_j \in \{0, 1\}$ and the embedding $x_j = e_j + v_j u_j$, with free position-indexed vectors e_j, u_j. A head computes $q_i = Qx_i$, $k_j = Kx_j$ with $Q, K \in \mathbb{R}^{d_h \times d}$, applies the unmasked row softmax $A(v)$ of $S(v)_{ij} = \langle q_i, k_j \rangle$ over the exact reals, and outputs $(A(v)v)_i$. The head is ε-*accurate* when $|(A(v)v)_i - \frac{1}{i}\sum_{j \leq i} v_j| \leq \varepsilon$ for every payload $v \in \{0, 1\}^n$ and every nonterminal row i. Write $B_\infty = \max_v B(S(v))$; the maximum includes $v = 0$. Every realized score matrix factors through the head, so $\rho(S(v)) \leq d_h$ for every v.

Theorem 10 (Adaptive extraction)**.** *Let a head be ε-accurate. For every row $i \geq 12$ with $i\varepsilon \leq 10^{-3}$,*

$$\max_j |A(0)_{ij} - T(i, j)| \leq 286\,\varepsilon,$$

and row i of $S(0)$ has a causal profile on every I with

$$\delta_i \leq 1144\,i\varepsilon, \qquad \gamma_i \geq \log \frac{1}{192\,i\varepsilon}.$$

The mechanism is exact rational algebra on one face. Fixing $v_i = 0$ freezes the query of row i; a payload bit then switches only its own key column, and the output on the payload $\mathbf{1}_X$ is an explicit odds expression in the $2(n-1)$ baseline and switched weights. Averages of the accuracy constraint over two central Hamming layers, with a leave-one-out difference, pin every prefix weight to $1/i$; switching every future bit on forces a sum of two nonnegative masses below 32ε, so future weight cannot hide in cancellation. Appendix E gives the proof.

For $0 < \varepsilon < 1$ and $1 \leq r$, let $B_r^{\mathrm{ad}}(n, \varepsilon)$ denote the infimum of B_∞ over ε-accurate heads with $d_h \leq r$, and $B_{r,p}^{\mathrm{ad}}(n) = B_r^{\mathrm{ad}}(n, n^{-p})$.

Corollary 11 (Adaptation does not change the law)**.** *For every fixed $p > 1$ and fixed positive integer r,*

$$\log B_{r,p}^{\mathrm{ad}}(n) = \frac{n-1}{r}\big((p-1)\log n + \log\log n + O_{p,r}(1)\big).$$

The lower half applies Theorem 3 to the single witness $S(0)$, whose centered rank is at most r; the Boolean class is used only to force the witness's profile. The upper half embeds the fixed construction of Theorem 4 adaptively with $u_j = 0$. Content adaptation through a linear payload channel therefore buys nothing at fixed rank: the two-term law is a property of the head, not of the fixed-matrix idealization.

Corollary 12 (Adaptive radius budget)**.** *Fix $p > 3/2$ and $c > 0$. An ε-accurate head with $\varepsilon = n^{-p}$ and $B_\infty \leq n^c$ has*

$$d_h \geq \left(\frac{p - 3/2}{c + p - 1} - o(1)\right) n.$$

Corollary 13 (Counting threshold)**.** *There are absolute constants such that every ε-accurate head with $\varepsilon < 1/(2n)$ satisfies $d_h = \Omega\big(\sqrt{n/(\log B_\infty + \log n)}\big)$; in particular $B_\infty \leq n^C$ forces $d_h = \Omega_C\big(\sqrt{n/\log n}\big)$.*

7 Related work

Dimension bottlenecks give the nearest architectural statements. Projection dimension at least the sequence length realizes an arbitrary positive column-stochastic pattern exactly, and below it some pattern fails [5]. Rank trades against head count for a nearest-neighbor target, exponentially in the rank deficit $d - r$ [4]. Patterns with at most k nonzero entries per row and column admit a ratio-preserving approximation at query/key dimension logarithmic in length, with a k^2 factor [10]. That guarantee constrains only ratios among entries the target leaves nonzero. Each charges a dimension; we solve for the magnitude a fixed dimension then requires, at $k = \Theta(n)$, for one named pattern.

One-layer lower bounds differ from ours in currency. Sparse averaging over input-specified positions charges embedding dimension m against bit precision p, with $mp = \Omega(q)$, by set disjointness [12]. Computing the frequency of the most frequent element charges $dhp = \Omega(\min\{m, n\})$ for dictionary size m [16]. Its counting construction sets the query matrix to zero—the score matrix is then input-independent—so that head is uniform over the whole context, with centered rank 0 and radius 0 here. Global averaging is free; the moving prefix boundary carries the entire price. Infinite precision removes bit counting, leaving embedding dimension and output-MLP size [8]. We charge logit magnitude under exact reals instead. Bounded entries also govern the computational side: subquadratic attention algorithms exist exactly when query/key entries stay $o(\sqrt{\log n})$ [1]; theirs bounds the time to compute a realized matrix, ours the magnitude required to realize one.

Bounded logit spread forces global softmax attention to disperse as sequence length grows [14]. The same work reports average spreads of 5.69 and 5.82 for Gemma 2B and 7B, with observed maxima 14.78 and 32.74; no rank enters. Lee studies row-centered logits—for an unmasked head, our SP_0—under the name energy field and reports concentration of their empirical singular spectrum [9]. Effective numerical rank requires a perturbation analysis before it can replace the exact rank in our theorem.

The one-sided determinant proof is related to Alon's two-sided perturbed-identity rank theorem [2], and to the ε-rank line that charges rank alone, with no magnitude budget [3]. The unnormalized prefix-sum matrix underlying T is factorized in the differentially private continual-observation line at a factor-norm cost logarithmic in the horizon, additively and without a rank constraint [6].

Construction-oriented treatments of transformers list prefix averaging as a basic ingredient, $(1/i)\sum_{k\le i} x_k$, obtained from the causal mask [15]. The formal-transducer line uses masked average-hard attention for the same prefix aggregation [13]. Neither charges a score magnitude for it. Within the fixed-matrix exact-real model of this paper, the law supplies the price of that ingredient once the mask is removed: at fixed centered rank r and worst-row error n^{-p}, the centered logit radius must reach $\exp(\Theta(n \log n/r))$, so a position-indexed score matrix that replaces an architectural support boundary pays a cost the masked realization does not. The statement concerns one fixed score matrix at one length, evaluated uniformly over bounded value streams; Section 6 extends the fixed-rank law to payload-adaptive scores, and no claim concerns trained heads.

8 Scope and open problems

The theorem concerns a fixed attention matrix that stays constant across bounded value probes. Taking $V = I_n$ reveals the matrix in one pass, while scalar probes give an equivalent test. Content-adaptive queries and keys need probe channels decoupled from their computation, for example value channels that the query and key projections ignore; Section 6 removes this requirement for Boolean payloads read through a linear channel. The model grants free position-indexed tables Q and K at each length. A shared positional generator and downstream correction fall beyond this matrix-level claim.

The rank charge covers the pure dot-product logits. An additive positional bias belongs in S when it is part of that score matrix. A separate T5-style bias can carry the full-rank step pattern from Proposition 8; its capacity falls beyond the d_h charge. An ALiBi-style decay also has its own positional channel, whose varying prefix scores alone fail to give uniform prefix weights.

Open questions: the behavior of Corollary 7 at the threshold $p \in (1, 3/2]$; the scale $\varepsilon \asymp 1/n$, where the two finite bounds' leading coefficients differ; a two-term law uniform in growing rank, where $-\log r$ meets $\log\log n$ (Appendix C); and a perturbation-stable form of (5) charging the singular values of SP_0 above a noise floor in place of exact rank, which would extend the audit of Section 5 to measured score matrices. The row-subset form of Theorem 3 also invites banded and sliding-window targets, whose profile geometry is not treated here; the growing-rank form of the adaptive comparison in Section 6 is open; and the permutation count of Lemma 14 is a natural next step for the Lean development.

A Output error and score geometry

Proof of Lemma 2. For $j \le i$, $T(i,j) = 1/i$, so entrywise error at most ε gives

$$\frac{1 - i\varepsilon}{i} \le A(i,j) \le \frac{1 + i\varepsilon}{i}.$$

Score differences are logarithms of probability ratios, so the oscillation on every subset of prefix columns is at most δ_i. For the adjacent key, $T(i,i+1)=0$ gives $A(i,i+1)\le\varepsilon$, while $A(i,i)\ge(1-i\varepsilon)/i$, which yields

$$S(i,i+1)-S(i,i)=\log\frac{A(i,i+1)}{A(i,i)}\le-\log\frac{1-i\varepsilon}{i\varepsilon}.$$

□

B Determinant proof

Lemma 14 (Permitted permutation count)**.** *For* $1\le k\le m$*,*

$$\#\{(\pi,K):K\subseteq[m],\ |K|=k,\ \pi\in\mathfrak{S}_m,\quad \pi(j)>j \text{ for all } j\notin K\}=k!\left\{{m \atop k}\right\}.$$

Proof. Fix K and write $[m]\setminus K=\{l_1<\cdots<l_{m-k}\}$. Assign images to $l_{m-k},\dots,l_1$ in that order. When l_i is reached, its admissible values $\{l_i+1,\dots,m\}$ number $m-l_i$. When $i<m-k$, all $m-k-i$ values already assigned exceed $l_{i+1}>l_i$; at $i=m-k$, no values have yet been assigned. Exactly $k+i-l_i$ choices remain. Since $l_i\ge i$ and $l_i\le k+i$, every factor lies in $\{0,\dots,k\}$. The k unused values go to K in $k!$ ways, so the count for fixed K is

$$k!\prod_{i=1}^{m-k}(k+i-l_i).$$

Put $a_i=l_i-i$, the number of elements of K below l_i. Then $K\mapsto(a_1\le\cdots\le a_{m-k})$ is a bijection onto weakly increasing sequences in $\{0,\dots,k\}$. Reversing the order and setting $i_j=k-a_{m-k+1-j}$ gives

$$\sum_K\prod_{i=1}^{m-k}(k+i-l_i)=\sum_{0\le i_1\le\cdots\le i_{m-k}\le k} i_1\cdots i_{m-k}=\sum_{1\le i_1\le\cdots\le i_{m-k}\le k} i_1\cdots i_{m-k}=h_{m-k}(1,2,\dots,k)=\left\{{m \atop k}\right\}.$$

Terms containing a zero factor vanish, which gives the second equality. The last identity follows from

$$h_d(1,\dots,k)=h_d(1,\dots,k-1)+k\,h_{d-1}(1,\dots,k).$$

With $d=m-k$, this is the Stirling recurrence $\left\{{m \atop k}\right\}=\left\{{m-1 \atop k-1}\right\}+k\left\{{m-1 \atop k}\right\}$. The initial values $h_0(1,\dots,k)=1$ and $h_d(1)=1$ match $\left\{{k \atop k}\right\}=1$ and $\left\{{m \atop 1}\right\}=1$. □

Proof of Theorem 3. Take the principal submatrix $D_I=D[I,I]$. Its rank is at most r. Let U_I be its upper-triangular part and $E_I=D_I-U_I$. Then

$$|\det U_I|\ge\prod_{i\in I}\gamma_i,\qquad U_I=D_I-E_I.$$

Expand the determinant multilinearly in columns. Index a term by $K\subseteq I$, the columns drawn from D_I; the remaining columns come from E_I. Terms with $|K|>r$ vanish. The last column of E_I is zero, so every surviving term has $|K|\ge1$.

Fix $|K|=k$. Order I increasingly and identify it with $[m]$. In the Leibniz expansion, a product can be nonzero when $\pi(j)>j$ for every $j\notin K$, because column j of E_I is supported strictly below row j. Lemma 14 counts the pairs (π,K) with this property.

Every product uses k entries bounded by $2B$. The other $m-k$ entries have distinct row indices and contribute at most

$$\max_{\substack{R\subseteq I\\|R|=k}}\prod_{i\in I\setminus R}\delta_i.$$

Summing absolute values proves (5); replacing all δ_i,γ_i by their common bounds proves (6). □

C Hadamard companion and all-rank floor

Proposition 15. *Under the constant-profile hypotheses of Theorem 3, with* $m=|I|$*,*

$$\gamma^m\le\sum_{k=1}^{\min\{r,m\}}e_{m-k}(1,\sqrt{2},\dots,\sqrt{m-1})\,\delta^{m-k}(2B\sqrt{m})^k,$$

where e_j is the elementary symmetric polynomial. Also

$$B \geq \gamma/2.$$

More generally, for row-dependent gaps in Theorem 3,

$$B \geq \frac{1}{2} \max_{i \in I} \gamma_i.$$

Proof. Use the same column expansion. A column j drawn from E_I has norm at most $\delta\sqrt{m-j}$, and a column drawn from D_I has norm at most $2B\sqrt{m}$. Hadamard's inequality and summation over the selected columns give the displayed elementary symmetric polynomial. Profiled row i contains two entries separated by at least γ_i; after centering, one has magnitude at least $\gamma_i/2$. Maximizing over the profiled rows gives the general floor, including its constant-profile specialization. □

Proof of Corollary 7. Take $I = [N]$ and use

$$e_{N-k}(1, \sqrt{2}, \ldots, \sqrt{N-1}) \leq \binom{N-1}{k-1} (\sqrt{N})^{N-k}.$$

Proposition 15 gives $2B \geq \gamma$; in the present output profile, $\gamma \geq \delta$ for all sufficiently large n. For every $k \leq r$,

$$(\delta\sqrt{N})^{N-k}(2B\sqrt{N})^k \leq (\delta\sqrt{N})^{N-r}(2B\sqrt{N})^r.$$

Summing the binomial coefficients yields

$$\gamma^N \leq 2^{N-1}(\delta\sqrt{N})^{N-r}(2B\sqrt{N})^r,$$

and rearrangement gives

$$N \log \frac{\gamma}{\delta\sqrt{N}} \leq (N-1)\log 2 + r \log \frac{2B}{\delta}.$$

At $\varepsilon = n^{-p}$, Lemma 2 gives $\gamma = (p-1)\log n\,(1+o(1))$ and $\delta = 2n^{1-p}(1+o(1))$, so the left side is $N((p-3/2)\log n + O(\log\log n))$. With $B \leq n^c$, $\log(2B/\delta) \leq (c+p-1)\log n + O(1)$. Dividing by $N \log n$ proves the corollary. □

D The upper construction

Proof of Theorem 4. For $1 \leq i \leq N$, set

$$\delta_i = \frac{i\varepsilon}{16}, \qquad \gamma_i = \log\frac{8(n-i)}{i\varepsilon}, \qquad w_i = \log\left(1 + \frac{\gamma_i}{\delta_i}\right).$$

When $0 < \varepsilon \leq 1/n$, their margins satisfy

$$\delta_i = \frac{i\varepsilon}{16} \leq \frac{1}{16}, \qquad \gamma_i = \log\frac{8(n-i)}{i\varepsilon} \geq \log 8, \qquad \frac{\gamma_i}{\delta_i} \geq 16\log 8 > 33, \tag{9}$$

using $i\varepsilon \leq N/n < 1$ and $n - i \geq 1$. These inequalities give $e^{\delta_i} \leq e^{1/16} < 2$ and $\gamma_i > \delta_i$.

Partition $[N]$ into exactly r consecutive nonempty blocks so that, for every $I = \{a, \ldots, b\}$,

$$\sum_{t=a+1}^{b} w_t \leq W/r, \qquad W = \sum_{i=1}^{N} w_i.$$

A greedy partition creates at most r blocks: at each cut, the completed internal load plus the skipped boundary weight exceeds W/r, and these charged sets are disjoint. If fewer blocks appear, $r \leq N$ guarantees a block with at least two rows. Splitting such blocks reaches exactly r and removes internal transitions.

For one block, define

$$c_a = 1, \qquad G_t = \sum_{s=a}^{t} \gamma_s c_s, \qquad c_t = \frac{G_{t-1}}{\delta_t} \quad (t > a),$$

$$u_i^I = \frac{\mathbf{1}[i \in I]}{c_i}, \qquad v_j^I = -\sum_{t=a}^{\min\{b,j-1\}} \gamma_t c_t, \qquad S = \sum_I u^I (v^I)^\top.$$

The u^I have disjoint nonempty supports. The first differences of the v^I have disjoint nonempty supports as well, so $\operatorname{rank}(S) = r$. Suppose $\sum_I a_I P_0 v^I = 0$. Then $\sum_I a_I v^I$ is constant. Taking first differences and using their disjoint nonzero supports gives $a_I = 0$ for every block, so $\rho(S) = r$.

By (9), $G_{t-1} \geq \gamma_a c_a = \gamma_a \geq \log 8$ and $\delta_t \leq 1/16$, so every $c_t \geq 1$.

For $i \in I = [a, b]$, the row has the explicit form

$$S(i,j) = \begin{cases} 0, & j \leq a, \\ -G_{j-1}/c_i, & a < j \leq b+1, \\ -G_b/c_i, & j > b+1. \end{cases}$$

Its prefix scores have maximum zero and range at most δ_i. The score drops by γ_i after key i. Let $\mathbf{1}_i$ denote the length-i all-ones vector, and let $z_j = -S(i,j) \in [0, \delta_i]$ for $j \leq i$. By (9),

$$G_t/G_{t-1} = 1 + \gamma_t/\delta_t \geq 2,$$

so geometric summation gives $\sum_{j \leq i} z_j \leq 2\delta_i$. Put $a_j = e^{-z_j}$, $Z = \sum_{j \leq i} a_j$, and let $\widehat{q}$ denote the attention distribution conditioned on the prefix. Since $Z \geq ie^{-\delta_i}$ and $1 - e^{-z_j} \leq z_j$,

$$\sum_{j \leq i} \left| \frac{a_j}{Z} - \frac{1}{i} \right| \leq \frac{2}{Z} \sum_{j \leq i} (1 - a_j) \leq \frac{2e^{\delta_i}}{i} \sum_{j \leq i} z_j.$$

Using $e^{\delta_i} < 2$ from (9) now gives

$$\left\| \widehat{q} - \frac{\mathbf{1}_i}{i} \right\|_1 \leq \frac{2e^{\delta_i}}{i} \sum_{j \leq i} z_j \leq \frac{4e^{\delta_i}\delta_i}{i} = \frac{e^{\delta_i}\varepsilon}{4} \leq \varepsilon/2.$$

Every future score is at most $-z_i - \gamma_i$, and each prefix score is at least $-z_i$. The future-to-prefix mass ratio bounds the future probability λ_i, giving

$$\lambda_i \leq \frac{n-i}{i} e^{-\gamma_i} = \varepsilon/8.$$

On the prefix, $A(i,:) = (1 - \lambda_i)\widehat{q}$. The triangle inequality gives full row error at most $\varepsilon/2 + 2\lambda_i \leq 3\varepsilon/4$. Row n is exact.

For a block,

$$G_b = \gamma_a \prod_{t=a+1}^{b} \left(1 + \frac{\gamma_t}{\delta_t}\right),$$

so $\log G_b \leq \log L_\varepsilon + W/r$, where $L_\varepsilon = \log(8n/\varepsilon)$. Since $\gamma_t \leq L_\varepsilon$, $\gamma_t/\delta_t > 1$, and $N! \geq (N/e)^N$,

$$w_t \leq \log \frac{32L_\varepsilon}{t\varepsilon},$$

$$W \leq N \log \frac{32L_\varepsilon}{\varepsilon} - \log N! \leq N \log \frac{32eL_\varepsilon}{N\varepsilon} \leq N \log \frac{64eL_\varepsilon}{n\varepsilon}.$$

Since every $c_i \geq 1$, row i of S has entries v^I_j/c_i of magnitude at most G_b. Its range bounds every centered entry, which gives (7). $\square$

E Content-adaptive extraction

Proof of Theorem 10. Fix a nonterminal row $i \geq 12$ and restrict to payloads with $v_i = 0$, so $q_i = Qe_i$ is fixed. Switching bit $j \neq i$ changes only key j. At the baseline $v = 0$, let

$$Z_i = \sum_t e^{S(0)_{it}}, \qquad p_j = \frac{e^{S(0)_{ij}}}{Z_i}, \qquad r_j = \frac{e^{S(e_j)_{ij}}}{Z_i} \quad (j \neq i),$$

and for $X \subseteq [n] \setminus \{i\}$ put $P_X = \sum_{j \in X} p_j$, $R_X = \sum_{j \in X} r_j$. On the payload $\mathbf{1}_X$, exactly the keys in X are switched and exactly they carry value one, so the output is

$$F_i(X) = \frac{R_X}{1 - P_X + R_X}, \tag{10}$$

and accuracy gives $|F_i(X) - |X \cap [i-1]|/i| \le \varepsilon$.

Let $J = [i-1]$, $k_1 = \lfloor i/3 \rfloor$, $k_2 = \lfloor 2i/3 \rfloor$, and $h_k = k/(i-k)$. The odds map $t \mapsto t/(1-t)$ sends $F_i(X)$ to $R_X/(1-P_X)$ and is 16-Lipschitz on $[0, 3/4]$; with $i\varepsilon \le 10^{-3}$ every target and value lies there for $k \le k_2$. Hence for every k-subset $X \subseteq J$, $k \le k_2$,

$$|R_X + h_k P_X - h_k| \le 16\varepsilon. \tag{11}$$

Let H be J or $J \setminus \{j\}$, $m_H = |H|$. Averaging (11) over all k_a-subsets of H replaces R_X, P_X by $\theta_a R_H, \theta_a P_H$ with $\theta_a = k_a/m_H \ge 1/4$, so

$$R_H + h_{k_a} P_H = \frac{m_H}{i - k_a} + E_a, \qquad |E_a| \le 64\varepsilon.$$

Since $\frac{m_H}{i}(1+h_k) = \frac{m_H}{i-k}$, subtracting the two equations and using $h_{k_2} - h_{k_1} \ge 9/10$ for $i \ge 12$ gives $|P_H - m_H/i| \le 143\varepsilon$. Applying this to $H = J$ and $H = J \setminus \{j\}$ and subtracting,

$$|p_j - 1/i| \le 286\varepsilon \qquad (j < i). \tag{12}$$

Let $L = \{i+1, \dots, n\}$. For a k_1-subset $Y \subseteq J$, the payloads $\mathbf{1}_Y$ and $\mathbf{1}_{Y \cup L}$ have the same prefix count, and subtracting their versions of (11) leaves $|R_L + h_{k_1} P_L| \le 32\varepsilon$. Both terms are nonnegative and $h_{k_1} \ge 1/3$, so

$$P_L \le 96\varepsilon, \qquad R_L \le 32\varepsilon. \tag{13}$$

Finally $p_i = 1 - P_J - P_L$ gives $|p_i - 1/i| \le 239\varepsilon$. Together with (12) and (13), every entry of row i of $A(0)$ is within 286ε of $T(i, \cdot)$.

Score differences are logarithms of probability ratios. With $286\, i\varepsilon \le 1/2$, the prefix oscillation is at most $\log \frac{1+286 i\varepsilon}{1-286 i\varepsilon} \le 1144\, i\varepsilon$, and $p_i \ge 1/(2i)$, $p_{i+1} \le 96\varepsilon$ give $S(0)_{i,i+1} - S(0)_{ii} \le \log(192\, i\varepsilon)$. ☐

Proof of Corollary 11. Set $\varepsilon = n^{-p}$. For all sufficiently large n, every row $i \in I = \{12, \dots, n-1\}$ satisfies the hypotheses of Theorem 10, giving on I the uniform margins $\delta = 1144\, n^{1-p}$ and $\gamma = (p-1)\log n - \log 192$. The witness $S(0)$ has $\rho(S(0)) \le r$ and $B(S(0)) \le B_\infty$, so Theorem 3 applies on I with $m = n - 12$ rows, and the algebra of Theorem 5 yields $\log B_\infty \ge \frac{m}{r}\big((p-1)\log n + \log\log n + O_{p,r}(1)\big) + O_p(\log n)$, which absorbs into the stated form. For the upper half, row-center and rank-factor the construction of Theorem 4, store its query and key factors in the e_j, set $u_j = 0$, and let Q, K select the blocks: a content-independent ε-accurate head with the same radius. ☐

Proof of Corollary 12. Set $\varepsilon = n^{-p}$. Theorem 10 gives the uniform margins $\delta = 1144\, n^{1-p}$ and $\gamma = (p-1)\log n - \log 192$ on $I = \{12, \dots, n-1\}$, and the witness satisfies $\rho(S(0)) \le d_h$ and $B(S(0)) \le B_\infty$. Proposition 15 and the proof of Corollary 7 run verbatim on I, with $m = n - 12$ in place of N: the rearranged inequality reads $m \log(\gamma/(\delta\sqrt{m})) \le (m-1)\log 2 + d_h \log(2B_\infty/\delta)$, whose left side is $m\big((p-3/2)\log n + O(\log\log n)\big)$ and whose right side is at most $(m-1)\log 2 + d_h\big((c+p-1)\log n + O(1)\big)$. Dividing by $m \log n$ and using $m = n(1 - o(1))$ proves the claim. ☐

Proof of Corollary 13. Put $r = d_h$, $K = 10^7$, $M = \lfloor n/(Kr) \rfloor$, and $I = \{\lceil M/2 \rceil, \dots, M\}$. If $M < 24$ then $r > n/(25K)$ and the claim is immediate. Otherwise every $i \in I$ has $i \ge 12$ and $i\varepsilon \le M/(2n) \le 10^{-3}$, so Theorem 10 gives uniform margins $\delta \le 572/(Kr)$ and $\gamma \ge \log(Kr/96)$ on I. If $r \ge |I|$ then $r = \Omega(\sqrt{n/K})$. Otherwise Theorem 3 on I with the estimates of Theorem 5 gives $\log B_\infty \ge c\, n/r^2 - \log r - C$, and combining the branches proves the statement. ☐

F Experimental details

The offline executable audit is `experiments/run_one_head_experiments.py`; it writes `experiments/output/one_head_bracket.csv`, `experiments/output/one_head_anchor.csv`, and `experiments/output/one_head_thresholds.csv`, together with `experiments/output/environment.txt`.

The manuscript source, the three audit programs, their committed outputs, and `C17/OneHead.lean` are available at https://anonymous.4open.science/r/rank-logit-law.

The bracket grid uses $n \in \{12, 16, 20, 24, 32, 40, 48, 56, 64\}$ and $r \in \{1, 2, 4, 8\}$. The program constructs rank-one block factors from binary64-generated budgets. High-precision decimal arithmetic recomputes each worst-row error and centered radius. The program also solves (5) and evaluates (7). Decimal precision follows the proved budget with 80 guard digits.

The threshold audit generates the representability table and the ambient dimension crossings at $n = 8192$. Its length searches record the last admitted value followed by the first excluded value. Dimension searches record the last excluded value followed by the first admitted value. The environment file identifies the interpreter, platform, Decimal exponent limit, and truncation cutoff used to produce the artifacts. Each search records its two-sided decision margin; the smallest reported is 2.3×10^{-4} nats. The companion program `experiments/certify_one_head_thresholds.py` recomputes every reported crossing and the $n = 8192$ anchor bound with exact integer Stirling numbers and outward-rounded interval arithmetic, certifying each decision in `experiments/output/one_head_interval_certificates.`

csv. The program `experiments/verify_finite_inequalities.py` checks the permutation count of Lemma 14 exhaustively through $m = 8$ and inequalities (5), (6), and the Hadamard companion directly on every bracket instance.

A Lean 4 development accompanies the project. For the statements of this paper it machine-checks the centered-rank-one, constant-profile specialization of Theorem 3: with the causal-difference profile taken as a hypothesis, $\gamma^N \le 2B\,\delta^{N-1}$ for every positive boundary count N (`C17/OneHead.lean`, building against Mathlib alone). The development builds with no `sorry` and no project axiom, and its theorems depend only on `propext`, `Classical.choice`, and `Quot.sound`. The general row-subset inequality, the permutation count of Lemma 14, the Hadamard companion, and the construction of Theorem 4 are not formalized; their finite-instance verification is the executable audit above.

References

[1] J. Alman and Z. Song. Fast attention requires bounded entries. *arXiv preprint arXiv:2302.13214*, 2023.

[2] N. Alon. Perturbed identity matrices have high rank: Proof and applications. *Combinatorics, Probability and Computing*, 18(1–2):3–15, 2009. doi: 10.1017/S0963548307008917.

[3] N. Alon, T. Lee, A. Shraibman, and S. Vempala. The approximate rank of a matrix and its algorithmic applications. In *Proceedings of the 45th Annual ACM Symposium on Theory of Computing*, 2013.

[4] N. Amsel, G. Yehudai, and J. Bruna. On the benefits of rank in attention layers. *arXiv preprint arXiv:2407.16153*, 2024.

[5] S. Bhojanapalli, C. Yun, A. S. Rawat, S. Reddi, and S. Kumar. Low-rank bottleneck in multi-head attention models. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 864–873, 2020.

[6] H. Fichtenberger, M. Henzinger, and J. Upadhyay. Constant matters: Fine-grained error bound on differentially private continual observation using completely bounded norms. *arXiv preprint arXiv:2202.11205*, 2022.

[7] Gemini Team, Google. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.

[8] A. Kozachinskiy. Lower bounds on transformers with infinite precision. *arXiv preprint arXiv:2412.20195*, 2024.

[9] W. Lee. On the invariants of softmax attention. *arXiv preprint arXiv:2605.02907*, 2026. URL https://arxiv.org/html/2605.02907v1.

[10] V. Likhosherstov, K. Choromanski, and A. Weller. On the expressive flexibility of self-attention matrices. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 8773–8781, 2023. doi: 10.1609/aaai.v37i7.26055.

[11] Llama Team, AI @ Meta. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

[12] C. Sanford, D. Hsu, and M. Telgarsky. Representational strengths and limitations of transformers. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[13] L. Strobl, D. Angluin, D. Chiang, J. Rawski, and A. Sabharwal. Transformers as transducers. *Transactions of the Association for Computational Linguistics*, 13:200–219, 2025. doi: 10.1162/tacl_a_00736.

[14] P. Veličković, C. Perivolaropoulos, F. Barbero, and R. Pascanu. Softmax is not enough (for sharp size generalisation). In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 2025.

[15] A. Yang, C. Watson, A. Xue, S. Bhattamishra, J. Llarena, W. Merrill, E. Dos Santos Ferreira, A. Svete, and D. Chiang. The transformer cookbook. *Transactions on Machine Learning Research*, 2026. URL https://openreview.net/forum?id=sPshCSvDrX.

[16] G. Yehudai, H. Kaplan, A. Ghandeharioun, M. Geva, and A. Globerson. When can transformers count to n? *arXiv preprint arXiv:2407.15160*, 2024.